

AgriLoan-Risk

A Data Science and Machine Learning Approach for Agricultural and Loan Risk Prediction

Research Paper

Abstract

Agricultural loans are closely tied to the success of farming. A farmer may have a stable financial profile when taking a loan, but a poor harvest can later make repayment difficult. This means that loan risk in agriculture is not only a financial issue. Weather, soil, pests, irrigation, and previous crop losses can also play an important role.

In this study, we developed AgriLoan-Risk as a data science and machine learning approach to examine this relationship. Our main idea was simple: instead of looking at financial information alone, we wanted to see whether climate and agricultural information could add useful signals for predicting crop loss and loan default.

We used a dataset containing 30,660 observations and studied two prediction tasks: crop-loss prediction and loan-default prediction. We designed three approaches. Approach A used financial features to create a loan-default baseline. Approach B used climate and agricultural features to predict crop loss. Approach C combined financial, climate, and agricultural features to predict loan default. For each approach, we tested Logistic Regression, Random Forest, XGBoost, and CatBoost, giving us 12 model experiments.

The results were encouraging. The best financial-only model, CatBoost, achieved 81.85% accuracy and 77.14% ROC-AUC. The selected integrated model, Logistic Regression, achieved 83.40% accuracy and 83.08% ROC-AUC. For crop-loss prediction, Logistic Regression achieved 80.14% ROC -AUC. Compared with the strongest financial-only baseline, the integrated approach improved ROCAUC by about 5.94 percentage points.

These results suggest that climate and agricultural information can add useful predictive information to financial loan-default modeling in our dataset. More importantly, the study shows how a real agricultural problem can be explored step by step using data preparation, feature design, machine learning, model comparison, and careful evaluation.

1.Introduction

Agricultural loans are different from many other types of loans because repayment is often connected to the success of a farming season. Farmers may borrow money for seeds, fertilizer, irrigation, labor, machinery, or other production costs. If the crop performs well, repayment may be easier. If the crop is damaged, the farmer's income can fall and repayment can become difficult.

This led us to a simple question: can agricultural loan risk be understood better if we look beyond financial information?

Traditional financial indicators such as income, loan amount, repayment history, and debt-to-income ratio are clearly important. However, they do not always tell us what may happen to agricultural production in the future. A farmer with a reasonable financial profile can still face serious problems after a weather event, crop failure, pest outbreak, or other production shock.

Our project therefore looks at agricultural risk and loan risk together. We use data science to test whether climate and agricultural information can improve the prediction of loan default when combined with financial information.

2.Problem Statement

The main problem is that agricultural production risk and financial loan risk are often considered separately. Financial data can tell us about a farmer's current financial position, while climate and agricultural data can tell us more about the conditions affecting production.

We wanted to know whether these two sides could be connected through machine learning. To study this, we first examined crop-loss prediction using climate and agricultural information. We then compared a financial-only loan-default model with an integrated model that also included climate and agricultural information.

3.Research Gap and Objective

The main gap we explored is the limited connection between agricultural production risk and financial loan-default risk within one data science study.

Our objective was not to assume that one particular model would solve the problem. Instead, we wanted to compare different information groups and several machine-learning models and let the results guide our conclusions.

The study focuses on four questions:

1. Can climate and agricultural information help predict crop loss?
2. How well can financial information predict loan default?
3. Does adding climate and agricultural information improve loan-default prediction?
4. Which of the tested models gives the strongest overall result?

4.Dataset

The dataset used in the project contains 30,660 observations. We worked with two main target variables: `crop_loss` and `loan_default`.

To make the experiments easier to understand, we organized the available predictors into four groups: climate, agriculture, farmer, and financial features.

Climate features describe weather and environmental conditions, including rainfall, humidity, temperature, sunshine, wind, pressure, and weather-stress measures.

Agricultural features include district, crop type, soil type, farm size, soil quality, pest risk, irrigation availability, crop insurance, previous crop loss, and farm vulnerability.

Farmer features include education level and farmer age. We kept these as a separate group during the project, but they were not included in the three main comparative approaches.

Financial features include annual income, loan amount, debt-to-income ratio, previous repayment score, household expense ratio, financial resilience score, and credit-risk index.

5.Data Preparation and Exploration

Before training the models, we first checked the structure of the dataset and examined data types, missing values, categorical variables, numerical variables, and the target columns. We then prepared the data for machine learning through encoding and numerical preprocessing.

We separated the target variables from the predictors and defined the feature groups before model training. A stratified train-test split was used so that the target distribution remained reasonably consistent between the training and test sets.

We also performed exploratory data analysis to understand the target distributions and the general behavior of the important variables. One important observation from the experiments was that some models could achieve good overall accuracy while still missing a noticeable number of positive cases. This is why we decided not to judge the models using accuracy alone.

6.Our Three Approaches

We designed three approaches so that we could answer the research questions in a clear and controlled way.

Approach A — Financial Baseline

Financial features → Loan Default

This was our starting point. It allowed us to see how well loan default could be predicted using financial information alone.

Approach B — Climate + Agriculture

Climate and agricultural features → Crop Loss

This approach helped us study whether production-related information could predict crop loss.

Approach C — Integrated Agricultural Risk

Financial + Climate + Agricultural features → Loan Default

This was the main experiment. Here, we combined the different types of information to see whether they could improve loan-default prediction.

For every approach, we tested four models: Logistic Regression, Random Forest, XGBoost, and CatBoost. In total, this gave us 12 experiments.

7. Machine Learning Models

We selected four commonly used models for tabular classification.

Logistic Regression gave us a simple and interpretable baseline. Random Forest allowed us to test a tree-based ensemble approach. XGBoost and CatBoost gave us two strong boosting-based alternatives.

Using several models was important because we did not want to assume in advance that a more complex model would always perform better. The experiments allowed us to compare the models using the same data and evaluation measures.

8. Model Evaluation

We evaluated the models using accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrices.

Accuracy tells us how many predictions were correct overall. Precision tells us how reliable the positive predictions were. Recall tells us how many of the actual positive cases were detected. F1score provides a balance between precision and recall. ROC-AUC helps us understand how well a model separates the two classes across different thresholds.

We used several metrics because agricultural risk prediction is not a problem where one number can tell the whole story.

9. Experimental Results

The results from all 12 experiments are shown in Table 1. The main pattern is that no single algorithm was best in every approach.

For the financial baseline, CatBoost produced the highest ROC-AUC at 77.14%.

For the climate-and-agriculture approach, Logistic Regression produced the highest ROC-AUC at 80.14%.

For the integrated approach, Logistic Regression produced the highest accuracy and ROC-AUC, reaching 83.40% accuracy and 83.08% ROC-AUC.

10. Financial Baseline vs Integrated Approach

The most important comparison in our study is between Approach A and Approach C because both are designed to predict loan default.

The best financial-only model was CatBoost:

Accuracy: 81.85%

F1-score: 35.33%

ROC-AUC: 77.14%

The selected integrated model was Logistic Regression:

Accuracy: 83.40%

F1-score: 46.76% ROC-AUC:

83.08%

The integrated approach therefore improved accuracy by about 1.55 percentage points, F1-score by about 11.43 percentage points, and ROC-AUC by about 5.94 percentage points.

This is the main result of the project. Within our dataset and experimental setup, adding climate and agricultural information gave the loan-default model a stronger overall ability to distinguish between the two classes.

11. Final Model Selection

Based on the experiments, we selected Logistic Regression as the final model for both prediction tasks.

For crop-loss prediction, the final model is Logistic Regression under Approach B, with an ROC-AUC of 80.14%.

For loan-default prediction, the final model is Logistic Regression under Approach C, with 83.40% accuracy and 83.08% ROC-AUC.

We selected these models based on the observed results and the balance between performance and interpretability. This does not mean that Logistic Regression will always be the best model for every agricultural dataset. It is the best choice among the models we tested in this study.

12. What We Learned from the Project

One of the most important lessons from the project was that a high accuracy score does not automatically mean that a model is performing well for every type of case. Some of our early results had reasonably high accuracy but weaker recall and F1-score for the positive class.

This pushed us to look at several metrics and to compare models more carefully.

We also learned that feature selection matters. Instead of mixing every available variable into one model from the beginning, we created clear feature groups and tested them through three approaches. This made it easier to understand what additional information contributed to the prediction.

Most importantly, the comparison between the financial baseline and the integrated approach gave us a measurable way to test our main research idea rather than simply assuming that climate and agricultural information would help.

13. Practical Meaning

The findings suggest that agricultural loan risk may be better understood when financial conditions are considered together with agricultural and climate conditions.

In a future real-world setting, this type of analysis could help identify potentially higher-risk cases earlier and provide additional information for human-led lending and risk-management decisions. For example, a borrower with moderate financial risk but unusually high production-related risk might deserve closer review.

However, our current study does not measure actual money saved or defaults prevented. Those outcomes would require testing the approach with real lending data and a properly designed realworld evaluation.

14. Limitations

We used real-world weather data in this study. However, we could not use actual bank loan records because borrower-level banking data are private and confidential. Therefore, the loan dataset was prepared for research purposes. If recent real-world loan data become available in future research, the models can be further tested and improved using actual banking records.

15. Future Work

There are several clear directions for improving the project.

First, the models can be tuned more systematically through hyperparameter optimization. Second, class imbalance and decision thresholds can be studied more carefully. Third, probability calibration and explainability methods such as SHAP can help make the predictions easier to understand.

The project can also be tested using temporal and cross-district validation, followed by external validation on larger real-world agricultural credit datasets. Future versions could include updated weather information and additional agricultural indicators.

Finally, a real-world pilot study would be needed to understand whether improved prediction actually leads to better lending decisions and measurable reductions in financial risk.

16.Conclusion

AgriLoan-Risk started with a simple question: can agricultural loan risk be understood better by looking at more than financial information?

To investigate this question, we built a complete data science workflow using 30,660 observations. We studied crop loss and loan default, created three experimental approaches, tested four machine-learning models, and compared 12 experiments using several evaluation metrics.

The strongest financial-only model achieved 77.14% ROC-AUC, while the selected integrated model achieved 83.08% ROC-AUC. This improvement suggests that climate and agricultural information can add useful predictive signals to financial loan-default prediction in our dataset.

The most important contribution of the project is therefore the evidence from the comparison itself. We did not simply assume that more information would produce a better model; we tested the idea and measured the difference. The results provide a useful starting point for future research using larger and more representative agricultural and credit datasets.

Table 1. Results of the 12 Model Experiments

Approach	Model	Accuracy	Precision	Recall	F1	ROC-AUC
A - Financial Only	Logistic Regression	0.8178	0.6199	0.2105	0.3143	0.7674
A - Financial Only	Random Forest	0.8076	0.5353	0.2245	0.3163	0.7497
A - Financial Only	XGBoost	0.8177	0.5961	0.2500	0.3523	0.7699
A - Financial Only	CatBoost	0.8185	0.6020	0.2500	0.3533	0.7714
B - Climate + Agriculture	Logistic Regression	0.8033	0.6823	0.4069	0.5098	0.8014
B - Climate + Agriculture	Random Forest	0.7921	0.6822	0.3232	0.4386	0.7848
B - Climate + Agriculture	XGBoost	0.7958	0.6618	0.3835	0.4856	0.7964
B - Climate + Agriculture	CatBoost	0.7996	0.6789	0.3842	0.4907	0.7988
C - Integrated Agricultural	Logistic Regression	0.8340	0.6422	0.3676	0.4676	0.8308

Risk						
C - Integrated Agricultural Risk	Random Forest	0.8280	0.6400	0.3026	0.4109	0.8185
C - Integrated Agricultural Risk	XGBoost	0.8315	0.6245	0.3775	0.4705	0.8240
C - Integrated Agricultural Risk	CatBoost	0.8332	0.6357	0.3717	0.4691	0.8285

Table 2. Main Comparison

Metric	Financial Baseline	Integrated Approach	Improvement
Accuracy	81.85%	83.40%	+1.55 pp
F1-score	35.33%	46.76%	+11.43 pp
ROC-AUC	77.14%	83.08%	+5.94 pp

Suggested Figures

Figure 1. AgriLoan-Risk data science workflow.

Figure 2. Crop-loss and loan-default target distributions.

Figure 3. ROC-AUC comparison across the 12 models.

Figure 4. Financial baseline versus integrated loan-default performance.

Figure 5. Conceptual relationship between climate/agricultural risk, crop loss, and loan-default risk.